Visualizing the Ramesside Star Clocks: Project Proposal

Jonah Walker, Consulted by Dr. Sarah Symons

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Background

The Ramesside Star Clocks is a collection of aincient Egyptian astronomical data painted into the tombs of Ramses the sixth, seventh, and eleventh in the Valley of the Kings in Luxor, Egypt. The data, which tracks the location in the night sky of 47 stars, is contained in a series of 24 tables, with data collection occurring periodically once every two weeks for a year (Neugebauer and Parker, 1960). These sorts of charts would have been transcribed by scribes into the tomb rooms from an original papyrus document. Unfortunately, that original document has been lost over time, as has any contextual writings or data surrounding the charts, including any information on the methods used to collect the data. The research focus of this lab is to determine those methods.

To do this, code that organizes fake night skies into the same format as the real star clocks (24 tables, with 13 rows representing hours and 7 columns representing sub-positions) will be used. This code can be adjusted to vary the methods used to collect the star data. Multiple iterations artificial night skies will be compared to the original aincient dataset, henceforth referred to as the E series. The R series is the iteration that is the fakest. It is an entirely procedurally generated set of 500 stars, all with different random magnitudes. There are five of them, named R1 through R5 respectively. The S series is based on the location of Sirius, all at the same magnitude as Sirius. Finally, the H series is the real Hipparcos catalogue, containing 517 stars up to magnitude 4.

For my IP, I will be using Python code to generate several data visualizations to reveal and compare trends in the series data. There are several requirements this algorithm must achieve. It must be easy and fast to use, as it is imperative other members of the lab can quickly and easily input generated datasets and compare it to the E series. It must be infinitely expandable, as the fake series could output any number of stars to the sets based on the collection method. The visualizations must be intuitive and nondeceptive such that they could appear in a scientific paper. The most important requirement is that it provides insight into how similar data collected from a fake night sky is to the real star clocks.

Charts

Generalized Table

One of the most complicated parts of this project is how many variables make up the star clocks. The stars themselves require unique identifiers. There are 47 in the E series, but some of the generated skies could be creating hundreds. Additionally, there are three dimensions of data associated with each star: the table(s) it was found in, the hour it appears in each table, and the position of the star on the table. As it is more than likely the magnitude of the star played a role in its selection for the star clocks, it will be accounted for as part of several visualizations. Each star must have an associated magnitude as an additional dimension of data.

The purpose of the generalized table is essentially a first stop for a fake chart coming into the visualization algorithm. It is much easier to create and adjust visualizations organized in the generalized table compared to how it is outputted from the star generation code. In this format, not only is the data more compact, but it is organized into one table rather than 24 of them, which is a lot easier to work with using the python libraries Pandas and Matplotlib.

Storing multiple dimensions of data in a table will require the use of sub columns. Because two of the three dimensions (hour and position) vary depending on the table the star is found in, the 'major' columns are table, with hour and position taking up sub columns. Assigning one row per star allows the list to be infinitely expandable, ideal for a fluctuating number of stars in each generated sky. One column will be associated with magnitude, which will remain independent from the sub columns because we assume the star will have the same magnitude in every table it appears in. If a star does not appear in a table, which is extremely common in both the E series and the fake series, the value in the sub columns will be NaN. Table 1 depicts an example generalized table.

Star	Magnitude	1	1	2	2	3	3	…
		Hr	Pos	Hr	Pos	Hr	Pos	
X000001	0.2	12	0	11	0	10	-1	
X000002	1.1	NaN	NaN	NaN	NaN	NaN	NaN	
X000003	3.9	NaN	NaN	11	-3	NaN	NaN	
X000004	-0.1	NaN	NaN	12	-2	11	-2	
X000005	-0.5	NaN	NaN	NaN	NaN	12	3	

Table 1: An example generalized table of the first 3 tables and first 5 stars in a hypothetical X series. The real chart will have 21 more columns (42 more sub columns) after column 3, and potentially hundreds more rows after X000005.

A benefit of the way the fake series of charts is coded is that each star is automatically assigned a unique identifier. The ID begins with the letter S, H, E, and R1 through R5 for the Sirius, Hipparcos, Egyptian, and 5 randomly generated series respectively. The letter is followed by six numerical digits unique to each star. My algorithm will automatically search through the 24 tables of the inputted series and populate the first column with the unique identifiers for each star. Because the position of one star does not affect the position of any other stars, the order the stars are populated into the first column does not matter.

Star Mapper

The purpose of the star mapping chart is to visualize the night sky that the code is generating. This is done using two maps: a 3D representation of the celestial sphere and a 2D circular stereographic projection. Both will be used so the user can determine whether the star generation code has created a fake night sky in an unwanted manner.

At the end of the project, if values of right ascension and declination can be determined for the E series, the code for the star mappers can easily be adjusted to show the locations of the stars the Egyptians used. This could very easily be overlayed against the

H series, which could be an excellent method to propose which stars the Egyptians were looking at and to determine whether Sirius in the E series is the same star we call Sirius today.

Star Stability

The purpose of the star stability chart is to track the tendency of stars to jump around between the seven column positions. This can be done using a 3D area graph, shown in Figure 1.

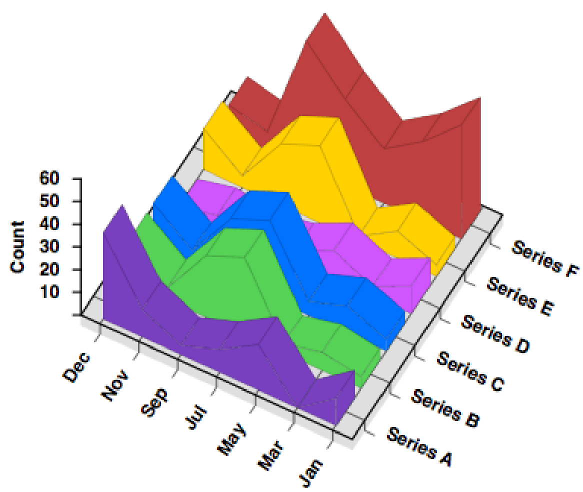


Image source

Figure 1: An example of a 3D area graph (Anon., 2017)

For our purposes, the count in Figure 1 will be replaced by position, ranging from negative three to three. The series will be replaced with individual stars, and the months will be replaced with star appearances. Star appearances is a new metric I will develop for this chart based on the table dimension. The greatest frequency a star is depicted in the E series is in 11 tables (there is one 12x-appearance star in the E series, but it appears in hour 12 twice, something that should not happen in the code). It will first appear in a table at hour 12, then in consecutively earlier hours over the course of the next ~11 tables. The first star appearance will be when the star enters its first table, typically in hour 12, the eleventh star appearance will be the last appearance of the star, typically in hour zero. The benefit of organizing the stars by star appearance is so we can directly compare how flat each area is next to each other, rather than trying to eyeball it across 24 tables where each star's 11 table streak does not align.

One potential issue with this visualization is low frequency stars appearing in less than a handful of tables, as well as stars with breaks that do not appear in consecutive tables. To effectively create this visualization, I will have to omit stars with low frequencies. When coding, it will be a judgment call on what the frequency requirement is, however, I estimate it will be around 7, which would qualify 22 of 47 stars in the E series.

Relating Hour Descent to a Slope

Where the previous visualization focuses in on the position dimension, this one focuses on the hour, specifically its tendency to decrease from 12 to zero through the tables. In this chart, hour will be plotted as a function of star appearance. Plotting hour versus star appearance for every star as a scatter plot would yield a line of best fit that is probably very close to $y = -x + 12$, where y is hour and x is star appearance. 12 being the initial value because stars tend to make their first appearance at hour 12. As a scatter plot, the slight fluctuations from this equation between different series might be useful, so such a scatter plot will be included in the 'low hanging fruit' dashboard. However, I believe a more interesting data visualization will be a comparison between the lines of best fit for each individual star.

As previously discussed, the ideal behaviour of a star is to appear in descending hours from 12 to zero over the course of 13 appearances, but this exact situation is not common in the E series. Usually, the star will skip hours along its descent such that it goes from hour 12 to zero over the course of eight to 11 appearances. This would yield a line of best fit with a slope closer to zero, approximately $y = -0.8x + 12$.

This visualization is going to communicate the rate of descent, as well as how similar the rates of descent are amongst the stars. Along with the graph, shown in Figure 2, this visualization will communicate the steepest slope, flattest slope, range of slopes, and average slope of the generated series and compare it to those from the E series.

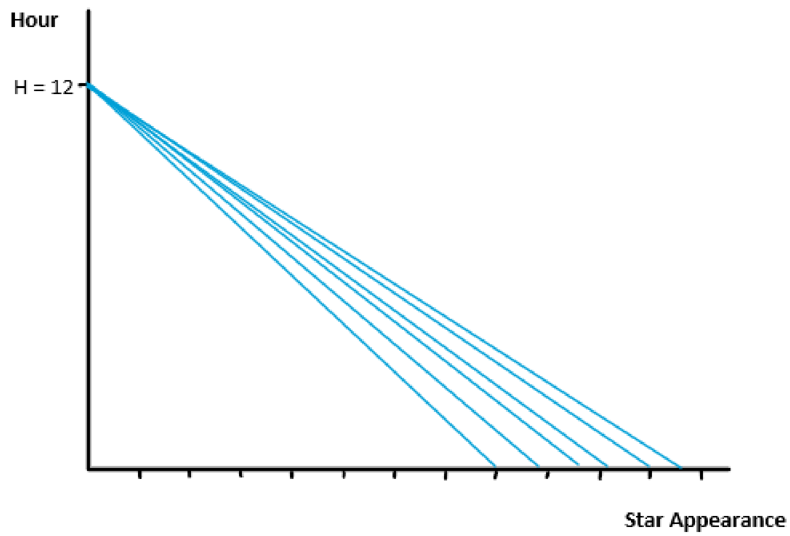
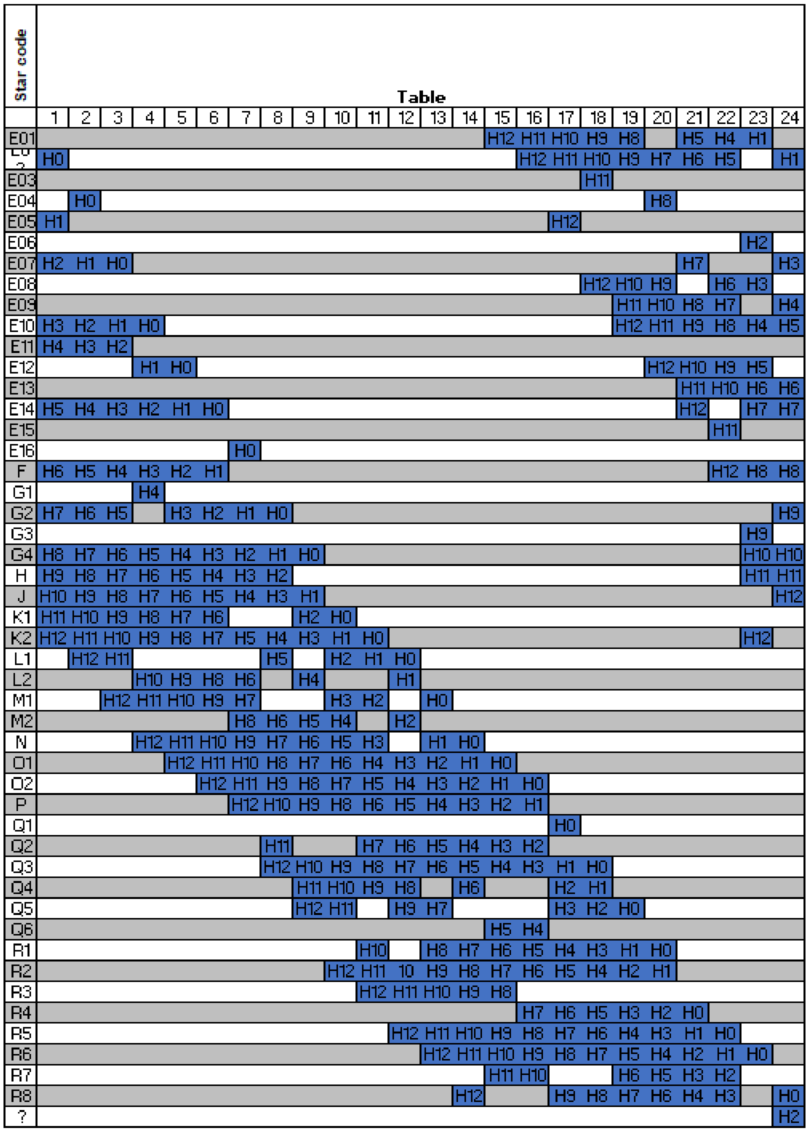


Figure 2: Proof-of-concept for the slope visualization. In the final version, the important slopes from E series stars (steepest, shallowest, average) can be projected here.

Frequency Table

The frequency table is a visualization that is very similar to a table found on the Star Appearances sheet of the RSC Curated Data Excel file, shown in Table 2. It has 24 columns, one for each table, and each row is one star. If the star is represented in the table, the cell will be filled in. The colour of the cell will depend on the hour, getting consecutively more transparent as the hours descend through the chart. I am going to organize it such that all the stars who make their first appearance in table one are in the first rows, followed by stars who make their first appearance in table two, and so on. The purpose of this visualization is to compare how filled in these charts become. I will also include an associated metric with this visualization, being % filled = number of filled cells / total number of cells * 100%.



Star code	Table																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
E01															H12	H11	H10	H9	H8		H5	H4	H1	
E02	H0															H12	H11	H10	H9	H7	H6	H5		H1
E03																		H11						
E04		H0																		H8				
E05	H1																H12							
E06																							H2	
E07	H2	H1	H0																		H7			H3
E08																		H12	H10	H9		H6	H3	
E09																			H11	H10	H8	H7		H4
E10	H3	H2	H1	H0															H12	H11	H9	H8	H4	H5
E11	H4	H3	H2																					
E12				H1	H0															H12	H10	H9	H5	
E13																					H11	H10	H6	H6
E14	H5	H4	H3	H2	H1	H0															H12		H7	H7
E15																						H11		
E16							H0																	
F	H6	H5	H4	H3	H2	H1																H12	H8	H8
G1				H4																				
G2	H7	H6	H5		H3	H2	H1	H0																H9
G3																							H9	
G4	H8	H7	H6	H5	H4	H3	H2	H1	H0														H10	H10
H	H9	H8	H7	H6	H5	H4	H3	H2															H11	H11
J	H10	H9	H8	H7	H6	H5	H4	H3	H1															H12
K1	H11	H10	H9	H8	H7	H6			H2	H0														
K2	H12	H11	H10	H9	H8	H7	H5	H4	H3	H1	H0												H12	
L1		H12	H11					H5		H2	H1	H0												
L2				H10	H9	H8	H6		H4			H1												
M1			H12	H11	H10	H9	H7			H3	H2		H0											
M2							H8	H6	H5	H4		H2												
N				H12	H11	H10	H9	H7	H6	H5	H3		H1	H0										
O1					H12	H11	H10	H8	H7	H6	H4	H3	H2	H1	H0									
O2						H12	H11	H9	H8	H7	H5	H4	H3	H2	H1	H0								
P							H12	H10	H9	H8	H6	H5	H4	H3	H2	H1								
Q1																	H0							
Q2								H11			H7	H6	H5	H4	H3	H2								
Q3								H12	H10	H9	H8	H7	H6	H5	H4	H3	H1	H0						
Q4									H11	H10	H9	H8		H6			H2	H1						
Q5									H12	H11		H9	H7				H3	H2	H0					
Q6															H5	H4								
R1											H10		H8	H7	H6	H5	H4	H3	H1	H0				
R2										H12	H11	10	H9	H8	H7	H6	H5	H4	H2	H1				
R3											H12	H11	H10	H9	H8									
R4																H7	H6	H5	H3	H2	H0			
R5												H12	H11	H10	H9	H8	H7	H6	H4	H3	H1	H0		
R6													H12	H11	H10	H9	H8	H7	H5	H4	H2	H1	H0	
R7															H11	H10			H6	H5	H3	H2		
R8														H12			H9	H8	H7	H6	H4	H3		H0
?																								H2

Table 2: The frequency table taken from the RSC Curated Data spreadsheet. In my visualization, instead of writing the hour on the cell like here, hour will be represented with a colour transparency spectrum.

All-Encompassing Perspective

The purpose of this chart is to put every dimension of data on a single chart in a 3D scatter plot. The x and y axes will be associated with table and position and the z axis will be associated with hour. The size of each dot will be related to its magnitude. The purpose of this chart is to provide a broad and all-encompassing look at the data and to check whether combining as many dimensions as possible yields any interesting spatial patterns.

Dashboard for 'Low Hanging Fruit'

The final visualizations will be included in a dashboard meant to showcase prominent patterns and metrics. They will be as follows:

- A bar chart comparing frequency and position meant to outline the heavy bias the E series has towards position zero and away from positions -3 and 3.
- A metric counting the number of unique stars. Any good result should have around 47 stars.
- A histogram of the total appearances of stars. Sorts the stars by how many times they appear in tables. In the E series, numerous stars only made one appearance in the tables, but only one that appeared in 12 tables.
- A scatter plot of hour as a function of star appearance. This was already discussed in the Relating Hour Descent to a Slope section.
- A table with the average and standard deviation of each star's position. Meant to be a numerical representation of star stability.
- A table showing how many stars are in each table.

All of these visualizations will be a comparison between the inputted dataset and the E series.

Works Cited

Anon. 2017. Further Exploration #1 3D Charts (Part 1). *The Data Visualisation Catalogue Blog*. Available at: <https://datavizcatalogue.com/blog/further-exploration-1-3d-charts/> [Accessed 15 February 2024].

Neugebauer, O. and Parker, R.A., 1960. *Egyptian Astronomical Texts*. Brown University Press.